

Computational Biofluid Mechanics

Patient-Specific Computational Fluid Dynamics of the Airways

Segmentation, Mesh Generation, and Flow Analysis in Tracheomalacia

Author: Henri Van Overmeire
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Repository: tracheomalacia_cfd

Template report—replace bracketed text and placeholder figures with final methods, results, and interpretations.

Abstract

Tracheomalacia is characterised by excessive collapse or narrowing of the tracheal lumen and can substantially alter respiratory airflow. This project uses patient-specific computed tomography, three-dimensional airway segmentation, and computational fluid dynamics (CFD) to compare airflow before and after intervention. Airway surfaces are segmented in 3D Slicer, clipped along VMTK-derived centerlines, extended in the local centerline direction, and closed with planar caps. Tetrahedral volume meshes are generated with Gmsh and steady incompressible airflow is solved using OpenFOAM. This report documents the reproducible processing pipeline, evaluates mesh quality and mesh independence, and compares clinically relevant flow quantities between the preoperative and postoperative geometries. [Complete this abstract with the principal numerical findings and conclusions.]

Contents

1	Introduction	1
1.1	Research questions	1
2	Critical review of a selected CFD paper	1
2.1	Goal	1
2.2	Methods	1
2.3	Results	2
2.4	Limitations	2
2.5	Evaluation	2
3	Imaging data and anatomical cases	2
4	Airway segmentation and surface preparation	3
4.1	Segmentation	3
4.2	Centerline extraction and clipping	3
4.3	Anatomical measurements and degree of constriction	3
5	Volume mesh generation	4
5.1	Near-wall resolution	5
6	Mathematical model	5
6.1	Governing equations	5
6.2	Boundary conditions	6
7	Numerical method	6
7.1	Steady solver	7
7.2	Transient solver and breathing waveform	7
8	Verification	7
8.1	Mesh quality	7
8.2	Convergence and mass conservation	8
8.3	Effect of additional steady iterations	8
8.4	Mesh-independence assessment	8
9	Results	9
9.1	Velocity field	9
9.2	Pressure distribution	9
9.3	Outlet and lung flow distribution	9
9.4	Local resistance before the bifurcation	11

10 Transient postoperative analysis	11
10.1 Time-step convergence	11
10.2 Velocity vectors at representative phases	11
10.3 Breathing waveform and time-varying resistance	13
11 Discussion	13
11.1 Mesh sensitivity and optimal mesh	13
11.2 Constant-diameter resistance during inspiration	13
11.3 Limitations	13
11.4 Handling transient non-convergence	14
12 Conclusions	14
Reproducibility statement	14

List of Figures

1	Airway segmentation and CFD-ready surface. Add representative preoperative and postoperative views with consistent orientation.	4
2	Measurement of constriction length and minimum lumen section in the preoperative geometry, with the corresponding postoperative section at the same anatomical location.	5
3	Computational meshes. Include a close-up in the stenotic region to demonstrate the change in resolution.	6
4	Residual evolution for the steady postoperative simulation. Mark the configured residual criteria and state whether monitored integral quantities were also stable.	8
5	Postoperative mesh-sensitivity study. The x-axis reports volume element count; each parameter and its percentage difference from the densest mesh are shown separately.	9
6	Velocity magnitude for the preoperative and postoperative cases. State the displayed time/iteration and use a shared colour range.	10
7	Static or kinematic pressure distribution. Clearly state whether pressure has been converted to pascals.	10
8	Fixed pre-bifurcation section used to calculate steady and transient local resistance. Plane origins and normals must be archived for reuse.	11
9	Residual evolution for the postoperative transient simulation. Highlight any time steps that did not satisfy the selected criterion.	12
10	Postoperative transient velocity vectors at three phases of the breathing cycle. State each time and inlet flow rate.	12
11	Breathing-cycle flow rate and local resistance. Resistance near zero flow must be handled carefully because direct division by $Q(t)$ becomes ill-conditioned.	13

List of Tables

1	Imaging and anatomical case information.	3
2	Final segmentation settings.	3
3	Anatomical measurements used to quantify the preoperative constriction. Replace placeholders with measurements from centerline-normal sections in 3D Slicer.	4
4	Replicable mesh settings and quality metrics. All current volume elements are first-order tetrahedra. Add the exact Gmsh version and final values from each <code>checkMesh</code> log.	5

5	OpenFOAM boundary conditions.	6
6	Software and provenance information. Replace placeholders with the immutable versions used for the final simulations.	7
7	Transient postoperative simulation settings.	7
8	Sensitivity of the selected axial-velocity monitor to 1000 additional steady iterations.	8
9	Mesh-independence comparison.	9
10	Outlet flow rates and fractions.	9
11	Postoperative left/right lung flow distribution.	10
12	Steady postoperative local-resistance calculation.	11

1 Introduction

Tracheomalacia involves reduced structural stability of the tracheal wall, which may cause dynamic narrowing and increased respiratory resistance. While medical imaging describes airway anatomy, CFD can quantify how geometric narrowing influences velocity, pressure loss, flow distribution, and local shear-related quantities.

The objectives of this project are to:

1. construct patient-specific preoperative and postoperative airway geometries from computed tomography;
2. establish a reproducible segmentation-to-simulation workflow;
3. quantify airflow and pressure differences between the two cases; and
4. assess whether the numerical conclusions are independent of mesh resolution.

1.1 Research questions

- How does the intervention change the minimum airway cross-section?
- How does the pressure drop across the trachea change?
- Where do peak velocities and recirculation regions occur?
- How is flow divided among the distal outlets?
- Are the reported quantities sufficiently mesh independent?

2 Critical review of a selected CFD paper

The selected paper is Taherian et al.'s patient-specific CFD evaluation of excessive dynamic airway collapse before and after tracheal stenting [1]. Each of the five required subsections below is shorter than 150 words. No paper figures are reproduced, remaining within the permitted maximum of three.

2.1 Goal

The study investigated whether patient-specific computational fluid dynamics could quantify the functional consequences of excessive dynamic airway collapse and assess improvement after tracheal stenting. The authors specifically compared inspiratory and expiratory airflow before and after intervention using CT-derived airway geometries. They aimed to determine how stenting affected pressure loss, velocity, wall shear stress, and turbulence, and whether these CFD quantities revealed clinically relevant changes that conventional pulmonary function tests might miss. The underlying question was therefore not merely whether the airway lumen became larger, but whether the anatomical intervention produced a meaningful improvement in respiratory flow behaviour.

2.2 Methods

Four patient-specific airway models represented inspiration and expiration before and after stenting. CT images were segmented in Mimics and simulated in STAR-CCM+. Each rigid model included the trachea and approximately six to eight airway generations. The authors simulated 2.5 s inspiration and 2.5 s expiration with sinusoidal resting-breathing flow. Patient-specific outlet pressure functions were derived in a preliminary step from lobar volume changes,

while the inlet was atmospheric with 10% turbulence intensity. Unsteady RANS with a low-Reynolds-number $k-\omega$ model and second-order spatial and temporal schemes was used. Inlet and outlet extensions measured one and four hydraulic diameters, respectively. Polyhedral meshes with prism layers were tested at 0.9, 1.7, and 3.2 million cells; 1.7 million cells was selected from pressure, velocity, and wall-shear-stress sensitivity.

2.3 Results

Stenting produced the clearest improvement during expiration. The severe pre-stent narrowing generated a high-velocity jet, recirculation, substantial pressure loss, and elevated wall shear stress. The reported pre-/post-stent expiratory pressure difference was approximately 26 cmH₂O, while mean tracheal wall shear stress decreased from 3.98 to 0.793 Pa, an approximately 80% reduction. Velocity variations also became smaller after stenting. Inspiratory changes were comparatively minor; mean wall shear stress increased from 0.0354 to 0.0813 Pa, and the stent introduced some local flow disturbance. Turbulence intensity increased in parts of both phases after intervention. Spirometry changed only modestly—FEV₁ increased by 290 mL—whereas CFD exposed pronounced expiratory functional improvement that pulmonary function testing did not clearly reflect.

2.4 Limitations

The study considered only one patient, so its quantitative findings cannot be generalised to all patients with excessive dynamic airway collapse. The airway walls were rigid even though wall deformation is the defining feature of the disease. Separate end-inspiration and end-expiration geometries represented the extremes of breathing rather than continuously moving anatomy; fluid–structure interaction was omitted because patient-specific tissue properties were unavailable. The upper airway and deeper peripheral generations were truncated, although both may influence tracheal flow and outlet impedance. Outlet pressures were inferred indirectly from lobar volume changes, and local CFD predictions were not validated against invasive pressure or velocity measurements. The RANS turbulence model and selected boundary conditions add model-form uncertainty. Finally, mesh sensitivity was demonstrated primarily for selected quantities in one pre-stent expiratory geometry.

2.5 Evaluation

For this patient, the authors largely answered their research question. The paired pre-/post-stent and inspiration/expiration design isolated the phase in which airway collapse mattered most, and pressure, velocity, and wall shear stress changed coherently after intervention. The contrast with relatively insensitive spirometry supports the claim that CFD can provide additional functional information. The mesh test and patient-informed outlet conditions strengthen the internal comparison. However, the study establishes a patient-specific proof of concept rather than a validated diagnostic or prognostic method. Rigid endpoint geometries cannot reproduce dynamic coupling between airflow and collapse, and a single case cannot establish clinical accuracy or treatment thresholds. Thus, the conclusion that stenting improved this patient’s expiratory airflow is supported, whereas broader claims about CFD-guided management require validation in more patients with dynamic or fluid–structure-interaction models.

3 Imaging data and anatomical cases

Two computed-tomography datasets are analysed: a preoperative case and a postoperative case. Table 1 should be completed from the DICOM metadata and segmentation logs.

Table 1: Imaging and anatomical case information.

Case	Voxel dimensions	Voxel spacing (mm)	Clinical state
Preoperative	[insert]	[insert]	Before intervention
Postoperative	[insert]	[insert]	After intervention

All patient-derived data must be handled according to the applicable privacy, ethics, and data-management requirements. No directly identifying information should appear in this report or repository.

4 Airway segmentation and surface preparation

4.1 Segmentation

The CT series is loaded into 3D Slicer. A case-specific seed is placed in the airway lumen, and connected region growth is performed within a selected Hounsfield-unit interval. Thresholds and seed coordinates are stored under `segmentation/assets/<case>/`. The result is inspected in axial, coronal, sagittal, and three-dimensional views to exclude lung parenchyma and external air while retaining the clinically relevant branches.

Record the final settings in Table 2.

Table 2: Final segmentation settings.

Case	Lower HU	Upper HU	Segmented voxels
Preoperative	[insert]	[insert]	[insert]
Postoperative	[insert]	[insert]	[insert]

4.2 Centerline extraction and clipping

A centerline network is calculated using VMTK. Visually refined clipping points are loaded as the Slicer markup node `AirwayCutEndpoints`. The airway is clipped using the same *Clip Vessel* logic as the SlicerVMTK user interface, with `AirwayNetworkModel` providing the centerline direction.

Flow extensions are aligned with the local centerline and adapt to the airway radius at each clipping location. Planar caps close the extensions for CFD. The final Slicer model is `AirwayExtendedSurfaceCapped` and is exported as `meshes/<case>/airways.stl`, preventing the two anatomies from silently overwriting one another.

4.3 Anatomical measurements and degree of constriction

The preoperative constriction is bounded by proximal and distal landmarks on the tracheal centerline. Its length is measured along that centerline. Cross-sections normal to the local centerline are sampled through the narrowed region to locate the minimum lumen area. Because the lumen need not be circular, an equivalent circular diameter is calculated from section area A as

$$D_{\text{eq}} = 2\sqrt{\frac{A}{\pi}}. \quad (1)$$

The postoperative area and equivalent diameter are measured at the matching anatomical location using the same procedure. The primary area-based degree of constriction is

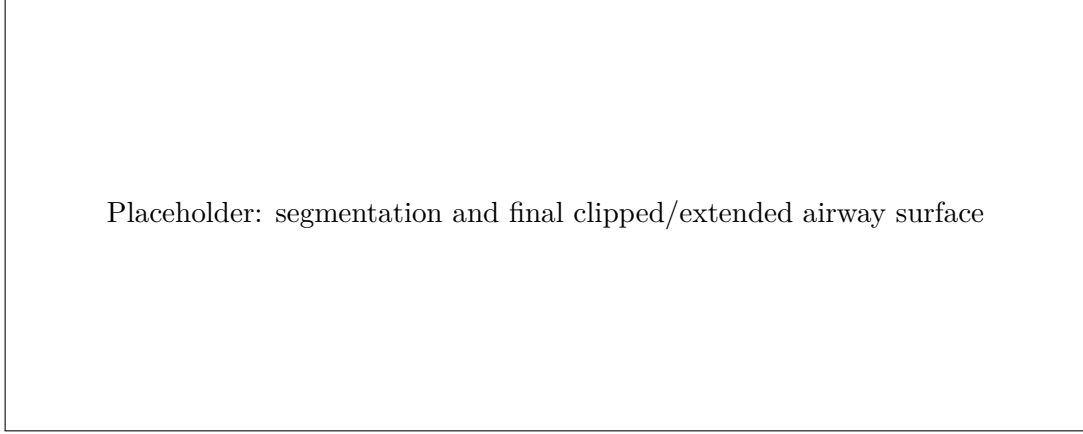


Figure 1: Airway segmentation and CFD-ready surface. Add representative preoperative and postoperative views with consistent orientation.

$$C_A = \left(1 - \frac{A_{\text{pre,min}}}{A_{\text{post,matched}}} \right) \times 100\%. \quad (2)$$

For transparency, the diameter reduction is also reported as

$$C_D = \left(1 - \frac{D_{\text{pre,min}}}{D_{\text{post,matched}}} \right) \times 100\%. \quad (3)$$

Table 3: Anatomical measurements used to quantify the preoperative constriction. Replace placeholders with measurements from centerline-normal sections in 3D Slicer.

Quantity	Symbol	Value
Preoperative constriction length	L_{pre}	[insert] mm
Preoperative minimum area	$A_{\text{pre,min}}$	[insert] mm ²
Preoperative equivalent diameter	$D_{\text{pre,min}}$	[insert] mm
Postoperative matched area	$A_{\text{post,matched}}$	[insert] mm ²
Postoperative matched diameter	$D_{\text{post,matched}}$	[insert] mm
Area constriction	C_A	[insert] %

5 Volume mesh generation

The triangulated STL surface is imported into Gmsh. Surface classification and geometry creation produce a closed fluid volume. The physical boundary groups are:

- **inlet**: tracheal inlet;
- **outlet_1**: upper-right outlet;
- **outlet_2**: lower-right outlet;
- **outlet_3**: left outlet;
- **wall**: remaining airway surface; and
- **fluid**: internal volume.

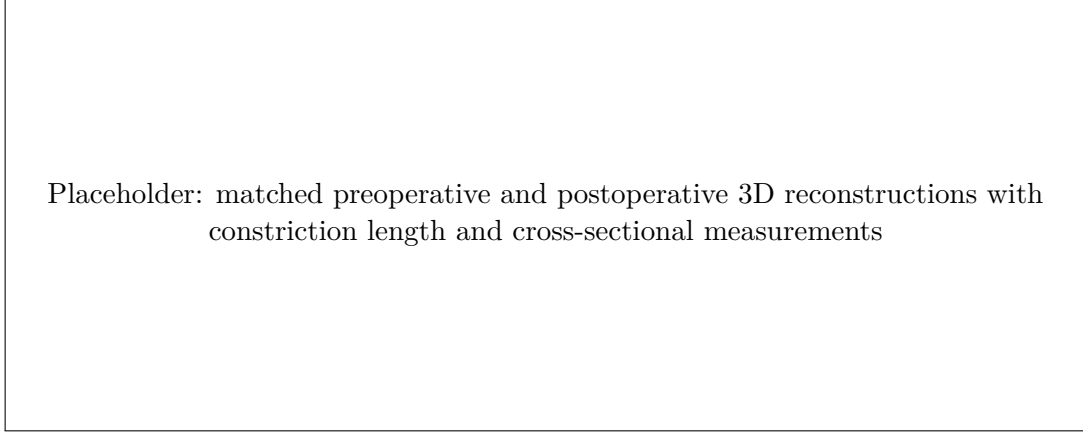


Figure 2: Measurement of constriction length and minimum lumen section in the preoperative geometry, with the corresponding postoperative section at the same anatomical location.

The physical groups are visually verified whenever the STL topology or Gmsh classification changes. The volume consists of first-order tetrahedra with a uniform target characteristic length controlled by `MESH_SIZE`.

Table 4: Replicable mesh settings and quality metrics. All current volume elements are first-order tetrahedra. Add the exact Gmsh version and final values from each `checkMesh` log.

Mesh	Target size (mm)	Points	Cells	Element type	Max. aspect ratio	Max. non-orthogonality
Coarse	[insert]	[insert]	[insert]	Tetrahedron	[insert]	[insert]
Baseline	0.25	[insert]	[insert]	Tetrahedron	[insert]	[insert]
Fine	0.15	[insert]	[insert]	Tetrahedron	[insert]	[insert]

5.1 Near-wall resolution

Boundary-layer prism elements are commonly used near no-slip walls because they resolve steep wall-normal velocity gradients efficiently and with better cell alignment than isotropic tetrahedra. The current Gmsh workflow uses only first-order tetrahedra and does not include dedicated inflation layers. This must not be presented as a hybrid boundary-layer mesh. Its consequences are considered when interpreting wall-adjacent quantities, and adding prism layers is identified as a model improvement.

The Gmsh mesh uses millimetres because it inherits the Slicer coordinate scale. After `gmshToFoam`, all coordinates are multiplied by 10^{-3} to obtain SI units.

6 Mathematical model

6.1 Governing equations

Air is modelled as an incompressible Newtonian fluid. For steady flow, mass and momentum conservation are:

$$\nabla \cdot \mathbf{u} = 0, \quad (4)$$

$$\rho(\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \mu \nabla^2 \mathbf{u}, \quad (5)$$

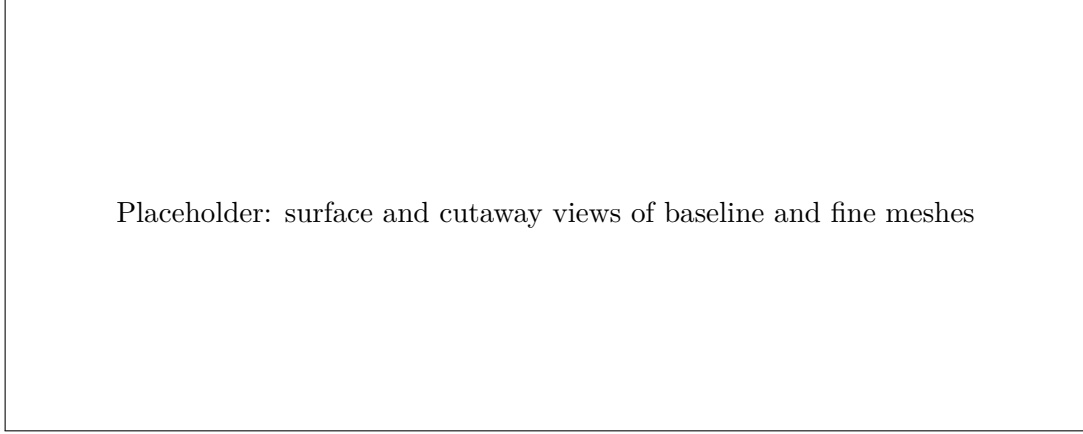


Figure 3: Computational meshes. Include a close-up in the stenotic region to demonstrate the change in resolution.

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and μ is dynamic viscosity. OpenFOAM uses kinematic pressure and the kinematic viscosity $\nu = \mu/\rho$. The configured value is $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$.

The current case uses the laminar model and the steady-state `simpleFoam` solver. The appropriateness of the laminar assumption should be assessed using a Reynolds number based on representative velocity U and diameter D :

$$\text{Re} = \frac{UD}{\nu}. \quad (6)$$

[Calculate Reynolds numbers at the inlet and stenosis and discuss whether a laminar, transitional, or turbulent model is justified.]

6.2 Boundary conditions

A prescribed volumetric flow rate is applied at the tracheal inlet:

$$Q_{\text{in}} = 1.6667 \times 10^{-4} \text{ m}^3/\text{s}. \quad (7)$$

This corresponds to approximately 10 L/min. A no-slip condition is applied at the airway wall. The three outlets use zero reference pressure with pressure-compatible velocity conditions. Table 5 summarises the implementation.

Table 5: OpenFOAM boundary conditions.

Patch	Velocity $\mathbf{u}$	Pressure p
inlet	flowRateInletVelocity	zeroGradient
outlet_1-3	pressureInletOutletVelocity	fixedValue 0
wall	noSlip	zeroGradient

7 Numerical method

The teaching staff approved the use of 3D Slicer and SlicerVMTK for geometry preparation, Gmsh for meshing, OpenFOAM for CFD, and independent local/remote compute infrastructure. This changes the software implementation but not the required physical analyses or validation.

Table 6: Software and provenance information. Replace placeholders with the immutable versions used for the final simulations.

Stage	Software	Version or identifier
Segmentation	3D Slicer + SlicerVMTK	5.12.2; extension [insert]
Surface/volume meshing	Gmsh	[insert]
CFD	OpenFOAM Docker image	2512; image ID [insert]
Post-processing	ParaView	[insert]
Source state	Git	commit [insert]

7.1 Steady solver

The finite-volume discretisation is implemented in OpenFOAM 2512. Pressure is solved using GAMG and velocity using a symmetric Gauss–Seidel smoothed solver. The SIMPLE algorithm uses two non-orthogonal correctors. The configured residual controls are 10^{-5} for pressure and 10^{-6} for velocity.

The parallel mesh is decomposed using the Scotch method. Simulations are run in the `opencfd/openfoam-default:latest` Docker image. The exact image ID and Git commit are recorded in Table 6.

7.2 Transient solver and breathing waveform

[Configure and document the Assignment 6 transient postoperative case. State the OpenFOAM solver, breathing-cycle duration, inlet flow-rate waveform, time-step size, maximum Courant number, number of inner correctors/iterations, write interval, number of simulated cycles, and criterion for cycle periodicity.]

Table 7: Transient postoperative simulation settings.

Parameter	Value
Solver	[insert]
Breathing-cycle duration	[insert] s
Time-step size	[insert] s
Maximum Courant number	[insert]
Simulated cycles	[insert]
Time-step convergence criterion	[insert]

8 Verification

8.1 Mesh quality

For each case and resolution, report the complete `checkMesh` summary. Particular attention should be paid to:

- maximum non-orthogonality;
- number of severely non-orthogonal faces;
- maximum skewness;
- minimum cell volume;

- maximum aspect ratio; and
- number of disconnected regions.

8.2 Convergence and mass conservation

Plot the pressure and velocity residual histories. Verify that the final outlet flow rates satisfy:

$$Q_{\text{in}} + \sum_{i=1}^3 Q_{\text{out},i} \approx 0, \quad (8)$$

using a consistent outward-normal sign convention. Report the relative mass imbalance.

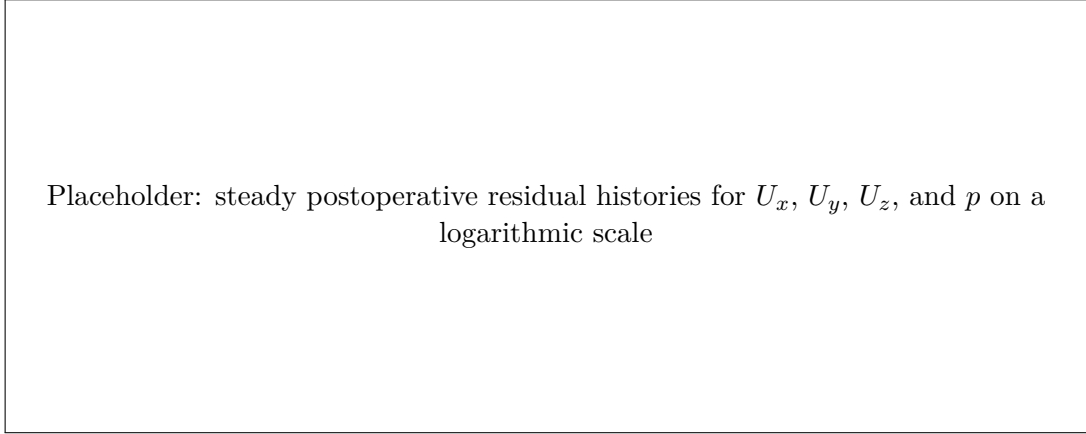


Figure 4: Residual evolution for the steady postoperative simulation. Mark the configured residual criteria and state whether monitored integral quantities were also stable.

8.3 Effect of additional steady iterations

Assignment 4 requests an assessment of whether 1000 additional iterations alter an axial-velocity parameter. A fixed centerline-normal section before the bifurcation is used, and its area-averaged axial velocity is recorded before and after continuation without reinitialising the solution.

Table 8: Sensitivity of the selected axial-velocity monitor to 1000 additional steady iterations.

Parameter	Iteration 2000	Iteration 3000	Relative change (%)
Area-averaged axial velocity	[insert]	[insert]	[insert]

8.4 Mesh-independence assessment

At least three postoperative mesh levels are preferred so that a trend can be assessed rather than only a pairwise difference. Each parameter of interest is plotted against the number of volume elements. For mesh i and the densest mesh d , the Assignment 5 percentage difference is defined as

$$\delta_{\phi,i} = \frac{|\phi_i - \phi_d|}{|\phi_d|} \times 100\%. \quad (9)$$

Table 9: Mesh-independence comparison.

Quantity	Coarse	Baseline	Fine/reference
Pre-bifurcation resistance	[insert]	[insert]	[insert]
Peak velocity	[insert]	[insert]	[insert]
Left-lung flow fraction	[insert]	[insert]	[insert]
Right-lung flow fraction	[insert]	[insert]	[insert]

Placeholder: one X–Y plot per parameter of interest versus mesh element count, plus percentage difference relative to the densest mesh

Figure 5: Postoperative mesh-sensitivity study. The x-axis reports volume element count; each parameter and its percentage difference from the densest mesh are shown separately.

9 Results

9.1 Velocity field

[Describe and compare the velocity magnitude, jet formation through stenotic regions, secondary flow, separation, and recirculation. Use identical colour scales for preoperative and postoperative figures.]

9.2 Pressure distribution

[Report pressure relative to the outlet reference and quantify pressure losses across anatomically relevant sections.]

9.3 Outlet and lung flow distribution

Outlet flow is obtained by integrating the normal volumetric flux over each outlet patch. The anatomically corresponding outlets are then grouped into left and right lungs. Report the grouping explicitly and use one outward-normal sign convention throughout.

Table 10: Outlet flow rates and fractions.

Case	Outlet 1	Outlet 2	Outlet 3	Mass imbalance
Preoperative	[insert]	[insert]	[insert]	[insert]
Postoperative	[insert]	[insert]	[insert]	[insert]

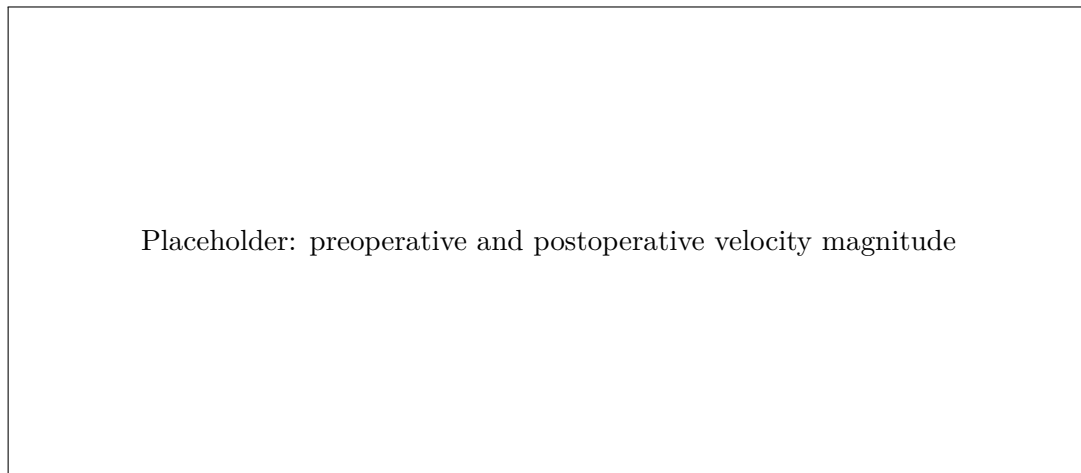


Figure 6: Velocity magnitude for the preoperative and postoperative cases. State the displayed time/iteration and use a shared colour range.

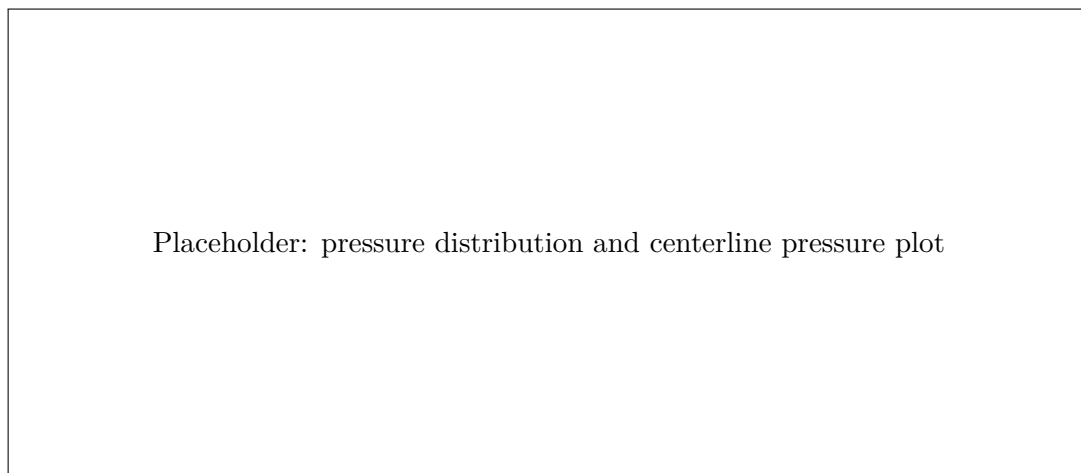


Figure 7: Static or kinematic pressure distribution. Clearly state whether pressure has been converted to pascals.

Table 11: Postoperative left/right lung flow distribution.

Lung	Flow rate (m ³ /s)	Fraction (%)
Left	[insert]	[insert]
Right	[insert]	[insert]

9.4 Local resistance before the bifurcation

Two planes normal to the centerline delimit a fixed section before the bifurcation, corresponding approximately to the location of the preoperative constriction. These same plane definitions are reused for every mesh and for the transient Assignment 6 analysis. With dimensional pressure difference ΔP and volumetric flow Q , local resistance is

$$R(t) = \frac{\overline{P}_{\text{up}}(t) - \overline{P}_{\text{down}}(t)}{Q(t)}. \quad (10)$$

For the incompressible OpenFOAM pressure field p , $\Delta P = \rho \Delta p$. Area-averaged pressure is used at each plane.

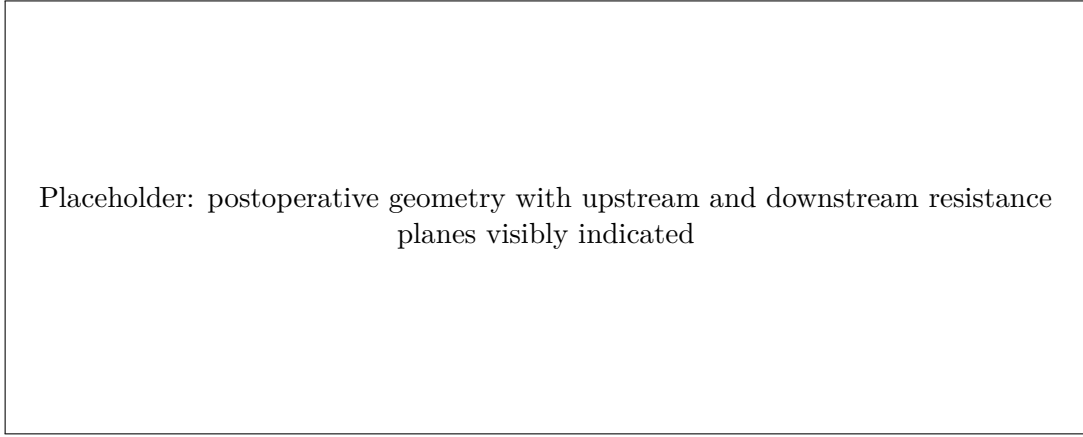


Figure 8: Fixed pre-bifurcation section used to calculate steady and transient local resistance. Plane origins and normals must be archived for reuse.

Table 12: Steady postoperative local-resistance calculation.

Quantity	Value
Upstream area-averaged pressure	[insert]
Downstream area-averaged pressure	[insert]
Pressure difference ΔP	[insert] Pa
Volumetric flow Q	[insert] m ³ /s
Resistance $R = \Delta P/Q$	[insert] Pa s/m ³

10 Transient postoperative analysis

10.1 Time-step convergence

[Plot residual evolution by time step, state the convergence criterion, and identify every time step that failed to meet it. Also report the maximum Courant number and whether the final simulated cycle is periodic.]

10.2 Velocity vectors at representative phases

Select three time points from the imposed breathing waveform, for example acceleration, peak inspiratory flow, and deceleration or flow reversal. The exact selection must be justified from the waveform rather than chosen only for visual appearance.

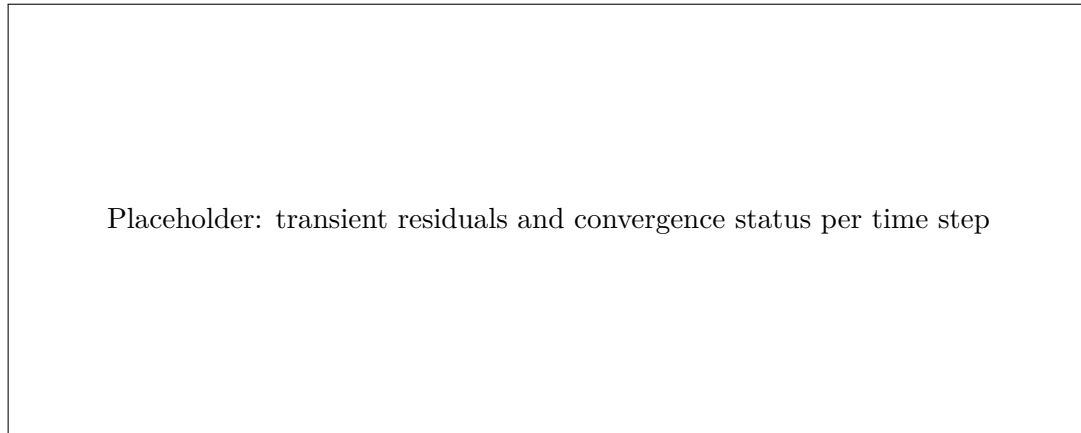


Figure 9: Residual evolution for the postoperative transient simulation. Highlight any time steps that did not satisfy the selected criterion.

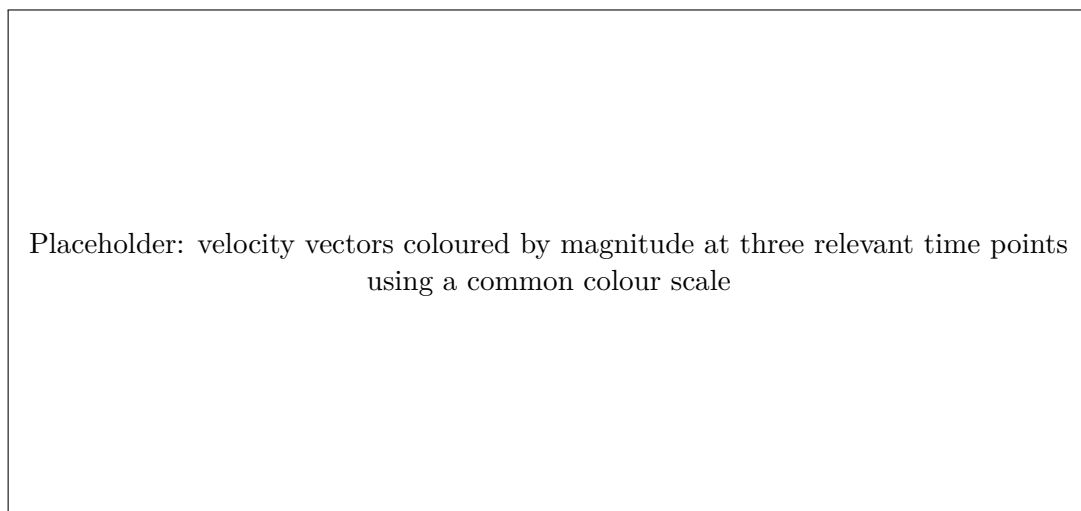


Figure 10: Postoperative transient velocity vectors at three phases of the breathing cycle. State each time and inlet flow rate.

10.3 Breathing waveform and time-varying resistance

The applied inlet flow rate and the local resistance from Equation 10 are evaluated over one complete breathing cycle. The computational postoperative resistance is compared with the constant-diameter expectation and with a qualitative preoperative sketch.

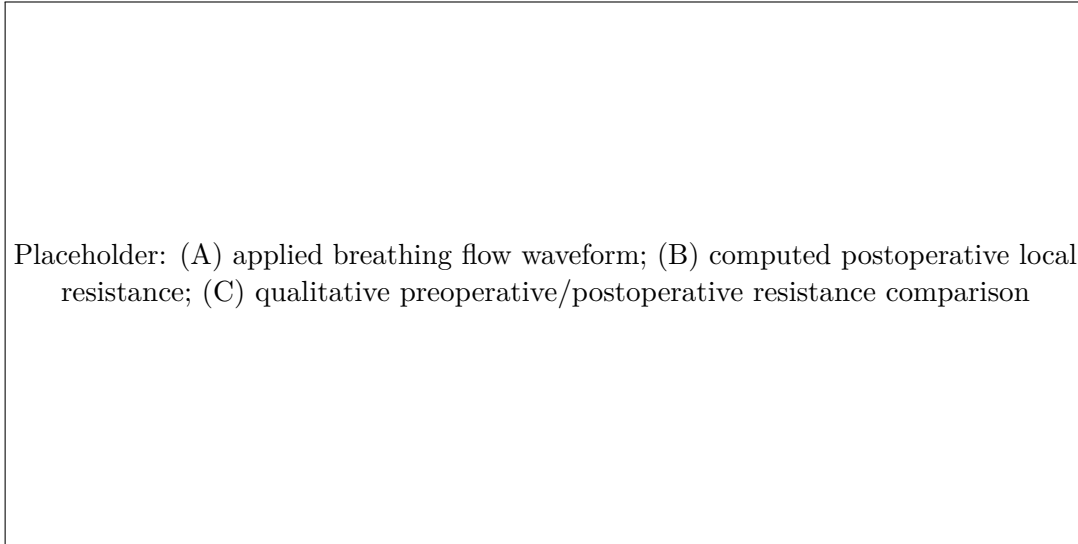


Figure 11: Breathing-cycle flow rate and local resistance. Resistance near zero flow must be handled carefully because direct division by $Q(t)$ becomes ill-conditioned.

11 Discussion

[Interpret the numerical results in relation to the anatomical differences and clinical question. Discuss whether pressure losses, velocity peaks, or outlet flow distributions improve after intervention. Distinguish numerical evidence from physiological inference.]

11.1 Mesh sensitivity and optimal mesh

[Interpret the parameter and percentage-difference plots in Figure 5. Select the coarsest mesh for which every clinically relevant integral quantity changes by less than the predefined acceptance criterion relative to the densest mesh. Justify that choice using both numerical accuracy and computational cost.]

11.2 Constant-diameter resistance during inspiration

[For fully developed laminar flow in a rigid constant-diameter tube, $\Delta P = RQ$ and the Poiseuille resistance is independent of flow rate. Therefore an increasing inspiratory flow produces a proportional pressure drop but constant $R = \Delta P/Q$. Discuss why developing flow, inertia, geometry, separation, or transition can make the computed airway resistance depart from this idealisation.]

11.3 Limitations

Potential limitations include:

- static CT geometry despite the dynamic nature of tracheomalacia;
- rigid airway walls and absence of fluid–structure interaction;

- steady inlet flow instead of a respiratory waveform;
- idealised outlet pressure conditions;
- uncertainty from segmentation and clipping-point placement;
- an isotropic first-order tetrahedral mesh without dedicated prism boundary layers;
- the selected laminar-flow assumption; and
- analysis of a limited number of anatomical cases.

With additional resources, the mesh study should include at least three systematically refined grids, local refinement around the stenosis and branch junctions, boundary-layer prism elements, and a formal convergence measure such as the grid-convergence index. A temporal-resolution study should likewise be performed for the transient model.

11.4 Handling transient non-convergence

[Identify unconverged time steps and discuss targeted remedies: reduce the time step, constrain the Courant number, increase inner iterations/correctors, improve mesh quality, use a better initial condition from a previous cycle, and simulate additional cycles until periodic behaviour is obtained. Report which changes would be tested first and why.]

12 Conclusions

[Summarise the principal quantitative findings, state whether the intervention improved the simulated flow characteristics, and report whether the conclusions were robust to mesh refinement.]

Reproducibility statement

The complete workflow is documented in `WORKFLOW.md`. Record the Git commit, Slicer version, Gmsh version, OpenFOAM image identifier, mesh sizes, cell counts, and solver settings used for the final results. The scripts `create_volume_mesh.sh`, `run_cfd.sh`, `run_fine_cfd.sh`, and `fetch_cfd_results.sh` automate the meshing, remote execution, reconstruction, and retrieval stages.

References

- [1] S. Taherian, H. Rahai, B. Gomez, T. Waddington, and F. Mazdisnian, “Computational fluid dynamics evaluation of excessive dynamic airway collapse,” *Clinical Biomechanics*, vol. 50, pp. 145–153, 2017. <https://doi.org/10.1016/j.clinbiomech.2017.10.018>.
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